

Beauty Books World

FAQ PDF Attachment Delivery Issue: Root Cause Analysis & Solutions

Component: FAQ Media & WhatsApp Dispatch

Service: ai-chat, process-message, send-whatsapp

Date: September 21, 2026

Status: Root Causes Identified & Verified

1. Issue Summary

Administrators configured 30+ comprehensive FAQs inside the dashboard, attaching official product guide PDFs (e.g., 1_whitening_treatment_face_cream.pdf) to each FAQ's `media_urls` field.

Observed Failure Behavior:

- When a customer asks questions answered by FAQs, the AI generates text, but **no PDF is sent** (`"faqMedia": []`).
- When a customer explicitly asks for a PDF, the AI rejects it:
"I do not have a PDF catalog available here, but I can provide all details directly! 🎯"

2. Local Environment Test Evidence

Running test queries locally against ai-chat confirmed the exact breakdown:

Test Case 1: Asking for a Product PDF Catalog

Inbound: "Can you send the PDF catalog for Advanced Facial Treatment Toners?"

```
{ "response": " ♦ Advanced Facial Treatment Toners\n✅ Formula 01: Korean Glass Skin Toner...\nWhich formula are you interested in? 🎯", "faqMedia": [] }
```

Result: AI responded with text, ignored the PDF request, and emitted no attachments.

3. Root Cause Analysis (4 Failure Points)

Point 1: AI Prompt Blindness (`media_urls` is Dropped)

In `supabase/functions/ai-chat/index.ts`, the `faqContext` string passed to the LLM only contains the question and text answers. Even though the database holds valid PDF URLs, the LLM has zero awareness that attachments exist.

Point 2: Contradictory Security Rules vs. Tracking Instructions

The system prompt enforces strict rules: *"NEVER show raw JSON, code, or technical markup"* alongside a secondary instruction: *"Include a <USED_FAQS> tag at the very end"*. Modern LLMs prioritize security rules, treating XML-like tags as technical markup and deliberately suppressing them.

Point 3: Information Duplication (Product Catalog vs. FAQs)

The system prompt contains both a *DYNAMIC PRODUCT CATALOG* and *FREQUENTLY ASKED QUESTIONS*. Step 4 instructs the AI to present formats and pricing directly from the catalog. Consequently, the AI thinks it's fulfilling requests using the catalog, bypassing FAQ citations entirely.

Point 4: Fragile Architecture (100% LLM-Dependent Dispatch)

The backend features zero deterministic fallback logic to check if a customer requested a file. It relies 100% on the probabilistic LLM outputting a specific XML tag. If omitted, the backend silently drops the attachment.

4. Comprehensive Solutions & Architecture Fixes

Solution A: Expose Attachments in System Prompt

Update `faqContext` to append metadata and clear instructions:

```
const hasPdf = Array.isArray(f.media_urls) && f.media_urls.some(u => /\.pdf/i.test(u));
const mediaNote = hasPdf ? `[ATTACHMENT: Official PDF Document available]` : "";
```

Solution B: Deterministic Backend Fallback (Smart Auto-Matching)

If the LLM omits the tag but query intent contains `pdf/catalog`, automatically match and attach the FAQ's PDF via backend logic.

Solution C: Storage URL Accessibility

Ensure internal Docker URLs (e.g., `http://172.17.0.1:8080/...`) are replaced with public MinIO / Storage endpoints so WAHA can fetch files successfully.

5. Comparison Table

Feature	Current State	Proposed Solution
AI PDF Awareness	Blind (no media info in prompt).	Fully Aware ([ATTACHMENT: PDF available]).
Response to "Send PDF"	Says: <i>"I do not have any PDF available"</i> .	Says: <i>"I am sharing the official guide PDF below"</i> .
Tag Generation	Dropped due to strict plain-text security rules.	Explicitly allowed and trained to append tags.
Backend Reliability	0% if LLM drops tag.	100% guaranteed via deterministic fallback.
WhatsApp Delivery	Empty (<code>faqMedia: []</code>).	PDF file delivered automatically with text.